Singapore Institute of Technology - University of Glasgow
Joint Degree in Computing Science Degree Programme

CSC3001 Capstone Project

Please complete the following form and attach it to the Capstone Report submitted.

Capstone Period: 29th August 2022 to 7th April 2023
Assessment Trimester: Start Trimester
Project Type: Industry

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I hereby acknowledge that I have engaged and discussed with my **Academic Supervisor** and **Work Supervisor** on the contents of this Capstone Report (Problem Definition) and have sought approval to release the report to the Singapore Institute of Technology and the University of Glasgow.

Signature

Date: 29th November 2022

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Singapore Institute of Technology - University of Glasgow
Joint Degree in Computing Science Degree Programme

Capstone Report (Problem Definition)

Dynamic Patrol Routing System

For **Start Trimester** from 29th August 2022 to 2nd December 2022

Student Name: Huang Jiashu
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Work Supervisor: *Kent Kok Tong Tan*

Academic Supervisor: Sye Loong Keoh

Submitted as part of the requirement for CSC3001 Capstone Project

1. *Introduction*

a. Project Motivation

Concorde Security Pte Ltd is a security service provider that offers a variety of innovative and cost-saving smart security services for the premises such as I-Guarding Solution, Intelligent Facility Authenticator, Intelligent Facility Sprinter and Robotics Technologies. The Company is recognized for its bold and disruptive innovation with its integrated monitoring of properties, assets and building service systems under 24/7 surveillance, ensuring complete security and business efficiency. This is done through a suite of smart security solutions called “I-Guarding Solutions”. The first of these solutions is its patented I-Man Facility Sprinter (“IFS”) which is a revolutionary mobile vehicular platform providing security and facility maintenance services.

Besides all the service that mentioned above, the company also provides daily night shift vehicle patrolling for each of the premises. Currently there are approximately fifty sites across Singapore that need to be visited every night and there are five vehicles in total that available for the patrol. However, the actual availability of vehicles would be determined by the health status of each vehicle. If any vehicle broke down or need to be sent for maintenance, the number of actual available vehicles will be lesser.

Before the vehicles going out for patrolling, the administrators in the command centre need to inform each vehicle drivers of the locations that they are supposed to visit every night. At each of the locations they have visited, the security guards are required to send images of the site to the administrators via WhatsApp just to confirm that they have visited the location. If any emergency alarms are triggered on any of the sites, the nearby vehicles would be notified by the administrators, and they are required to present at the location.

However, after the vehicle check up on the site and ready to continue patrolling, they need to return to the original route and carry on from what is left which is not very time-efficient and fuel-efficient since more fuel would been wasted if they are required drive back to the previous site instead of continuing from the current location.

Therefore, it is necessary to come up with a scalable system that not only able to auto plan the route for each vehicle every day but also can re-route the patrolling route for those vehicles in order to reduce the carbon footprint emission.

b. Problem Definition

According to a report posted on Impactful, one gallon of gasoline emits around 8857 grams of C02 when combusted [1]. Being one of the greatest factors of global climate change, carbon emission can lead to a variety of adverse environmental impacts such as rising sea level, changing precipitation patterns, ocean acidification and so on [2]. Therefore, immediate action should be taken to reduce the carbon emission to mitigate climate change. Additionally, one of the global goals is to reach “zero emissions” by 2050, which means that there would be no newly added emissions to the atmosphere [3]. Therefore, this project is potentially contributed to achieve this global goal. There are four types of vehicle routing problems categorised by information quality and evolution, which is static and deterministic, static and stochastic, dynamic and deterministic, dynamic and stochastic [4]. Since all the sites are known but random sites can have emergency so the problem I try to solve will be static and stochastic. Some of the vehicle routing problem optimization techniques include meta-heuristics, for example, Genetic Algorithm (GA), Ant Colony System (AS), Multi-Particle Swarm Optimization (MAPSO), and Tabu Search (TS) [5]. CLARITY (Clustering, eLbow-method, polAR-routing, InTeger-programing and evolutionarY) metho also can be used for solving such issues [6]. Besides this, this project also aims to targets some of these following problems.

The current patrol route planning is manually done by the administrators at the command centre, so they have to plan for each vehicle every day. But right now, there are only around fifty sites and five vehicles which is still manageable. However, when the company's business grows bigger, manual planning will become less scalable and therefore a dynamic planning system could be a much preferrable way.

When the security vehicle reaches at each location, they need to take a picture of the premises and send it to the administrators via WhatsApp. But the admins can lose the pictures very easily, but this can be solved by integrating a database to organize all the data. A centralized database management system can improve data integrity, reduce data redundancy and so on. My proposed solution would include a centralized database to store all the pictures sent by the security guards for administrators to view and

This project is aimed to target at the current global climate change issues at a bigger picture, but it also targets at relatively small problem such as eliminate manpower needed to be involved in the planning of the patrol routing and store data in a more centralized system.

In conclusion, this Capstone Project is aimed to design an automatic patrol routing system which would take in daily available vehicles and locations needed to be visited and utilize algorithms to plan for the possible shortest routing distance for each vehicle. Besides this, re-plan the route for vehicles who get involved in any site incidents are also required in order to maximize time and fuel efficiency which can potentially result with carbon footprint emission reduction.

2. ***Project Objectives***

a. High Level Target

- This project is aimed to effectively reduce the carbon emission based on the difference of milage covered by the original route and the proposed solution.
- This project is also aimed to automate the patrol route planning system and eliminate the needs for the administrators in the command centre to plan the patrolling route for each vehicle every night.

b. Project Objectives

- The dynamic patrol routing system would plan out the patrol route for each vehicle every day depends on the number of sites and vehicles available every day. It would also ensure all the sites are covered and the overall patrolling distance is the shortest.
- The dynamic patrol routing system would be able to automatically re-plan the route for those vehicles that involved in the emergency alert triggered at the site. With this system being implemented, those vehicles no need to drive extra miles which could significantly reduce the carbon footprint emission.
- The system would include a dashboard at the command centre for administrators use. The information shown on the dashboard would include data such as real-time location tracking of each vehicle, details of the vehicle on duty (car plate number, driver in charge, vehicle health status, milage coverage), emergency alert noticeboard, estimated carbon footprint emission and so on.
- A mobile application would also be implemented for the security guards on the vehicle to use for them to view the plan for the patrol every night and they are also able to contact the admins directly via this mobile application.

3. References

[1] G. Smoot, "What is the carbon footprint of gasoline? here are the facts," Impactful Ninja. [Online]. Available: https://impactful.ninja/the-carbon-footprint-of-gasoline/#:~:text=On%20a%20life%2Dcycle%20basis,directly%20contributes%20to%20climate%20change. [Accessed: 29-Nov-2022].

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[4] a review of dynamic vehicle routing problems - researchgate. (n.d.). Retrieved November 30, 2022, from https://www.researchgate.net/publication/257196300_A_review_of_dynamic_vehicle_routing_problems

[5] Kucharska, E. (2019, April 15). *Dynamic vehicle routing problem-predictive and unexpected customer availability*. MDPI. Retrieved November 30, 2022, from https://www.mdpi.com/2073-8994/11/4/546/htm

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